Theorising Service Routines through Ontological Granularity and Behavioural Sequencing

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Abstract

Digital trace data offer unprecedented access to service encounters, yet translating them into mechanism-based theory remains difficult. Drawing on critical realism and behavioural operations management, this paper introduces ontological granularity tuning as a method for moving between observable digital residues and generative routines. Using movement-tracking logs from 2,871 exhibition visitors (90,420 zone-visits), we operationalise five behavioural blocks and apply computational sequence analysis and K-Means clustering. We identify four stable routine archetypes, with a dominant "high-activity exploratory-browsing" routine anchored by spatial affordances, reframing service-routine theorising as a retroductive, granularity-sensitive process.

Keywords: Behavioural sequencing, Service routines, Behavioural routines, Critical realism

Introduction

Digital trace data have fundamentally transformed the capacity of operations researchers to observe human activity within service environments, yet a substantial gap remains between high-resolution digital residues and the generative mechanisms that drive operational routines. Behavioural operations management (BOM) has long called for theory grounded in the cognitive and behavioural micro-foundations of service processes (Bendoly et al., 2006; Gino and Pisano, 2008), but computational approaches tend to privilege data-driven discovery over explanation, producing theory-light analytics that lack conceptual depth (Berente et al., 2019). Moving from description to explanation therefore requires a disciplined way of linking observable digital patterns to the underlying routines that produce them.

This paper develops a critical realist framework for theorising service routines from digital trace data. Specifically, we introduce the concept of ontological granularity

tuning—the systematic variation of the theoretical resolution at which behavioural blocks are defined—as a methodological lever for identifying semi-stable demi-regularities across levels of abstraction. We operationalise the framework through computational sequence analysis (CSA) and K-Means clustering applied to a large-scale movement-tracking dataset from an experiential service (an industry exhibition), and we examine the stability of four candidate routine archetypes, with particular attention to the theorised Re-engagement Loop as a non-linear return pattern.

The research addresses two questions. First, how can digital trace data be translated into mechanism-based routine theory without collapsing into descriptive analytics? Second, which routine archetypes emerge, and how stable are they across different ontological resolutions? The paper contributes: (i) a critical realist method for computationally intensive theorising of service routines; (ii) empirical evidence on the stability of a dominant exploratory-browsing routine in a real service setting; and (iii) managerial implications for the design of service spaces that channel visitors between routine states.

Theoretical background

Service routines and behavioural operations

Routines are repetitive, recognisable patterns of interdependent actions that provide the operational backbone of service delivery (Pentland and Feldman, 2008). In experiential services, such as exhibitions, museums and retail environments, routines are jointly produced by actors and by the affordances of the physical setting (Zomerdijk and Voss, 2010; Johnston, 1999). Behavioural operations research has shown that aggregate performance outcomes are shaped by micro-level cognitive and behavioural regularities (Bendoly et al., 2006; Gino and Pisano, 2008), yet most empirical work has relied on laboratory experiments or coarse-grained surveys. Digital trace data—logs, RFID traces, sensor streams—now allow direct observation of these micro-foundations at scale, but on their own they do not furnish theory.

Critical realism and computationally intensive theorising

Critical realism distinguishes three strata of reality: the empirical (observed digital traces), the actual (behavioural events as they occur), and the real (generative mechanisms that produce events; Bhaskar, 1978; Mingers et al., 2013). Theorising is retroductive: inferring from stable empirical patterns the mechanisms that, if they existed, would best explain what is observed (Wynn and Williams, 2012). Berente et al. (2019) argue that computationally intensive theorising succeeds when computational procedures are embedded within this retroductive logic, and Zhang et al. (2022) show how computational sequence analysis can operationalise retroduction by detecting semi-stable patterns—demi-regularities—in behavioural strings.

Ontological granularity tuning

We build on these foundations by introducing ontological granularity tuning: a deliberate movement between coarse and fine representations of behaviour. At a very fine resolution, traces appear as idiosyncratic noise; at a very coarse resolution, they collapse into a single "visit". Between these extremes lie levels of abstraction at which stable patterns—routines—become visible. Tuning granularity is therefore a theoretical operation: the analyst iteratively re-codes digital atoms into behavioural blocks of differing generality and inspects which level yields the most stable, theoretically meaningful demi-regularities.

Applied to experiential services, we define five behavioural blocks that vary in cognitive commitment and temporal structure: Exploration (EX) is low-commitment scanning, typically short dwell at a node; Focal Browsing (FO) is evaluative attention of moderate length; Deep Engagement (DE) captures sustained cognitive absorption; Re-engagement (RE) is a purposeful return to a previously visited node after a gap of at least 180 seconds; and Transition (TR) is short movement between nodes. Higher-order routines are motifs in strings over this alphabet. The Re-engagement Loop—a non-linear pattern in which an actor returns to earlier nodes—is of particular theoretical interest because, if stable across demographic subgroups, it points to a mechanism driven by spatial affordances rather than by individual differences (Chase and Dasu, 2001).

Research design and methodology

Setting and data

The empirical setting is a large-scale business-to-business exhibition held over three days in an enclosed venue with 15 thematic zones. Visitor movement was captured via RFID-tagged passes registering entry and exit at each zone. After pre-processing—including removal of invalid records, merging of consecutive visits with gaps of five seconds or less, and derivation of per-visit features—the working dataset contains 90,420 zone-visit records generated by 2,871 unique visitors. Mean per-visit duration was 73.3 seconds (median 25 s); mean total visits per user were 31.5 (median 23) and mean total dwell time per user was 2,307.6 seconds (median 1,447 s). Age band, gender, visitor type and device operating system were retained as metadata.

From digital atoms to behavioural blocks

Each zone-visit was algorithmically translated into one of the five behavioural blocks described above. Duration thresholds were derived from the empirical distribution of dwell time: visits below the 33rd percentile (16 seconds) were flagged as EX candidates; intermediate durations as FO; and sustained visits in the upper tail as DE. Revisits to the same zone separated by more than 180 seconds were coded as RE. Short inter-zone intervals served as TR markers. Each user's sequence of blocks constitutes a behavioural string that is analogous to a genetic sequence (van der Aalst, 2016; Zhang et al., 2022).

Computational sequence analysis and clustering

We proceeded in three analytical stages. First, exploratory analysis characterised the empirical distribution of the atoms (durations, inter-visit gaps, zone popularity, hourly traffic). Second, we engineered thirteen sequence-level features, including block ratios (RE, DE, EX, FO, TR), motif ratios (Loop_Ratio = RE→RE; Move_Return = TR→RE; Deep_Return = DE→TR→RE; Scan_Return = EX/FO→TR→RE), structural indices (Sequence_Length, Unique_States, Transition_Density) and a Revisit_Balance index (RE − (EX+FO)). Third, we applied K-Means clustering after standardisation and examined four candidate granularities (k = 2, 3, 4, 5). Cluster quality was assessed jointly by the Elbow method and silhouette scores (Table 1), and solutions were visualised through PCA projection plus standardised feature heatmaps. This pipeline enacts ontological granularity tuning in two senses: vertically (across the hierarchy of atoms → blocks → motifs → routines) and horizontally (across candidate cluster numbers).

Table 1 – Cluster validity across candidate granularities (k = 2–8)

k	Inertia (SSE)	Silhouette	Dominant cluster (%)	Notes

2	14,943.2	0.461	81.8%	Over-aggregates routines
3	11,571.2	0.353	54.3%	Unstable boundary
4	8,806.8	0.414	54.1%	Preferred solution
5	7,393.7	0.416	53.5%	Splits short-visit type
6	6,022.0	0.363	—	Silhouette drops
7	5,211.7	0.353	—	Unstable
8	4,553.2	0.362	—	Over-refined

Findings

Empirical stratum: distribution of behavioural atoms

Figure 1 summarises the empirical stratum. The overall duration distribution (panel A) is strongly right-skewed: a third of all zone-visits last 16 seconds or less and the median is 25 seconds, indicating a dominant scanning regime at the atomic level. Panel B disaggregates by visit phase and shows that first-visit, middle-visit and last-visit duration distributions are substantially overlapping, while last visits carry a heavier tail (mean 138.2 s vs. 70.4 s for middle visits), consistent with end-of-visit deep engagement. The inter-zone gap distribution (panel C) exhibits a clear exponential decay with a median of 19 seconds, suggesting that transitions are predominantly short and local. Panel D, however, reveals only a weak Pearson correlation ($r = -0.01$) between gap length and zone popularity, indicating that longer gaps are not systematically driven by the attractiveness of individual zones.

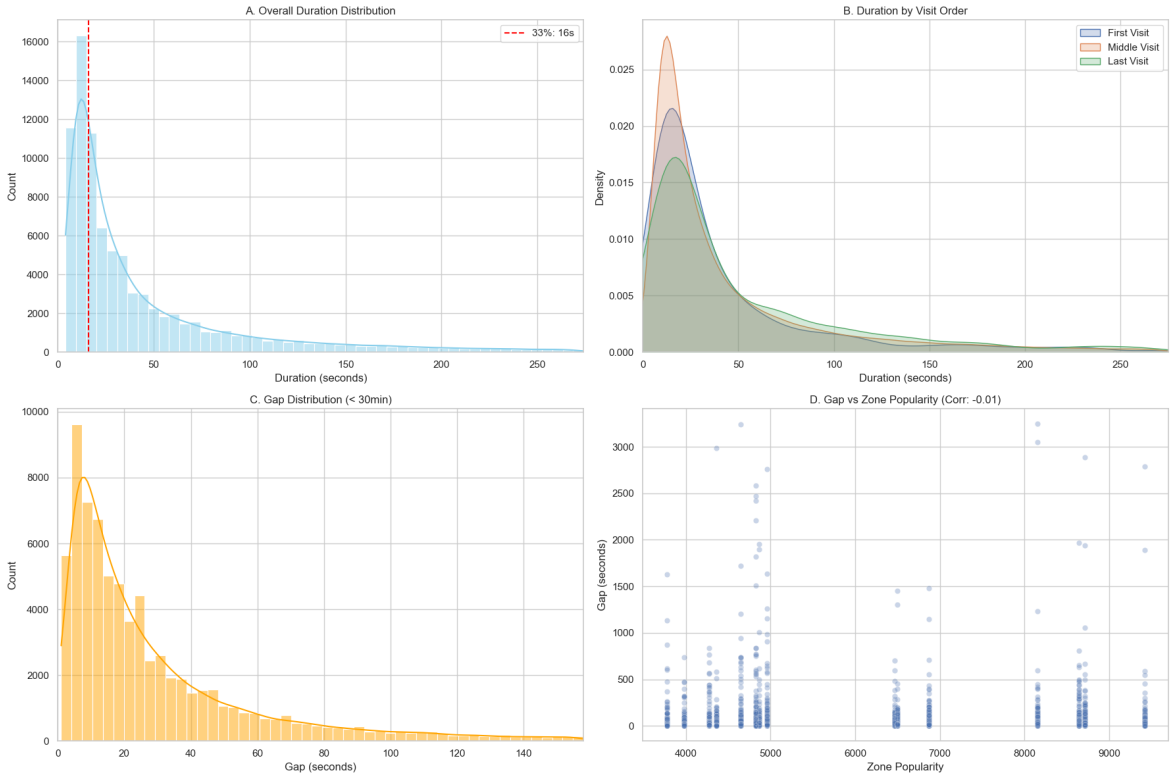


Figure 1 – Empirical distribution of behavioural atoms: (A) duration, (B) duration by visit phase, (C) inter-zone gaps, (D) gap vs. zone popularity

Hourly traffic (Figure 2) is bimodal with pronounced peaks at 14:00–15:00 (c. 16,000 visits per hour) and a secondary peak around 11:00. When visit durations are examined at the zone level (Figure 3), three zones—zone 9, 12 and 6—consistently stand out with higher median dwell (48, 40 and 36 seconds respectively) and heavier upper tails. These "slow" zones are empirically where DE-type atoms concentrate and where the preconditions for Re-engagement Loops are most likely to be satisfied.

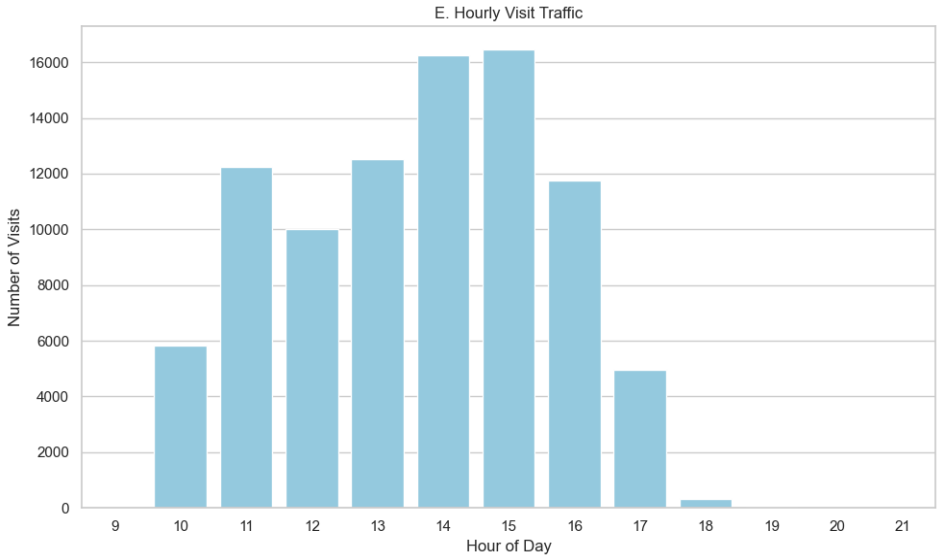


Figure 2 – Hourly visit traffic over the exhibition period

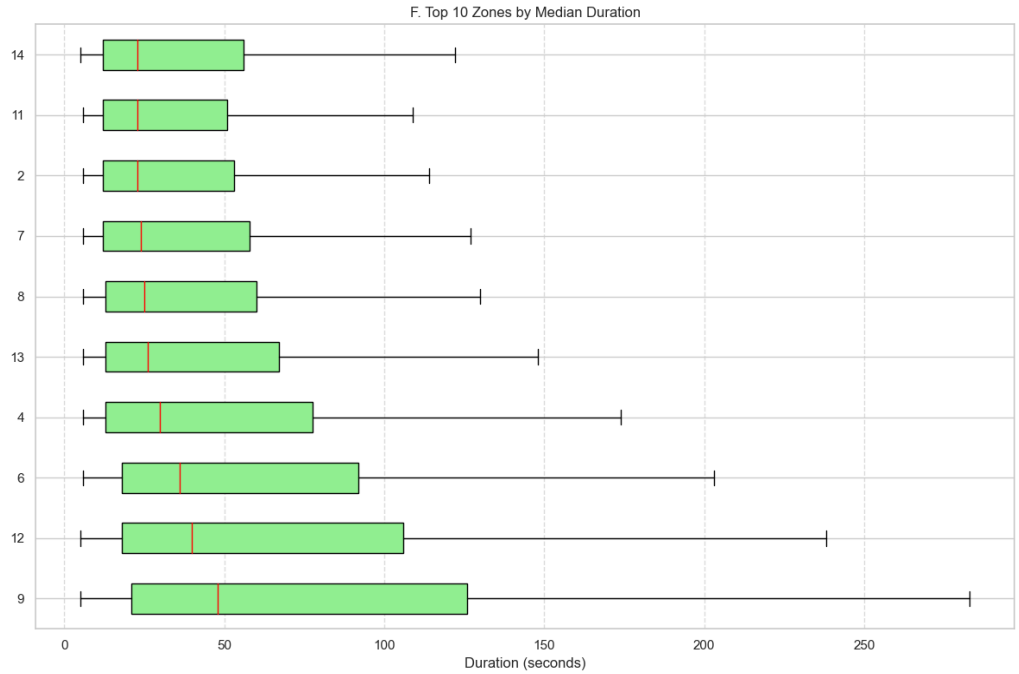


Figure 3 – Top 10 zones by median visit duration

Actual stratum: behavioural events and user-level structure

At the user level, the median visitor traverses 22 zones over 1,329 seconds of dwell time. A right-skewed tail of highly active users contributes the bulk of long sequences. These totals already foreshadow two candidate routines: a large group of users who browse

many zones lightly, and a smaller group who engage deeply with fewer zones. Table 2 reports cluster-level means for the selected four-cluster solution and translates them into qualitative routine archetypes using the descriptors generated by the automated interpretation routine.

Table 2 – Four routine archetypes from K-Means clustering (k = 4)

Archetype	N (%)	Seq. length	DE ratio	EX ratio	FO ratio	Unique states	Rev. balance
High-activity explorer	1,636 (54.1%)	91.3	0.08	0.12	0.12	8.3	−0.24
Mid-range browser	828 (27.4%)	20.2	0.05	0.19	0.21	5.7	−0.39
Pass-through	326 (10.8%)	1.0	0.00	0.00	0.00	1.0	−0.00
Focused deep-engager	232 (7.7%)	7.1	0.35	0.03	0.06	4.0	−0.09

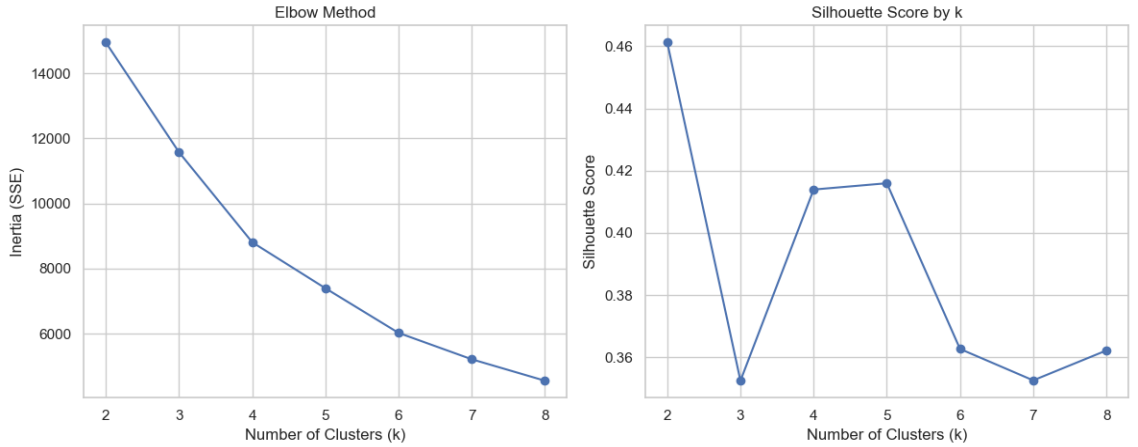


Figure 4 – Cluster validity: elbow (inertia) and silhouette score by k

Figure 4 supports the choice of k = 4: silhouette recovers to 0.414 after a dip at k = 3 and inertia exhibits a visible bend at k = 4. Figure 5 visualises the four archetypes in the reduced PCA space and reports the standardised feature profile of each cluster. The High-activity explorer cluster dominates (54.1% of users), combining the longest sequences, the highest number of unique zones and moderate levels of scanning (EX) and focal browsing (FO). The Mid-range browser captures shorter exploratory sequences dominated by scanning. The Focused deep-engager cluster—small but distinctive—is marked by a four-to-seven-fold relative intensity in DE. The Pass-through cluster registers essentially a single event and represents non-participating traces.

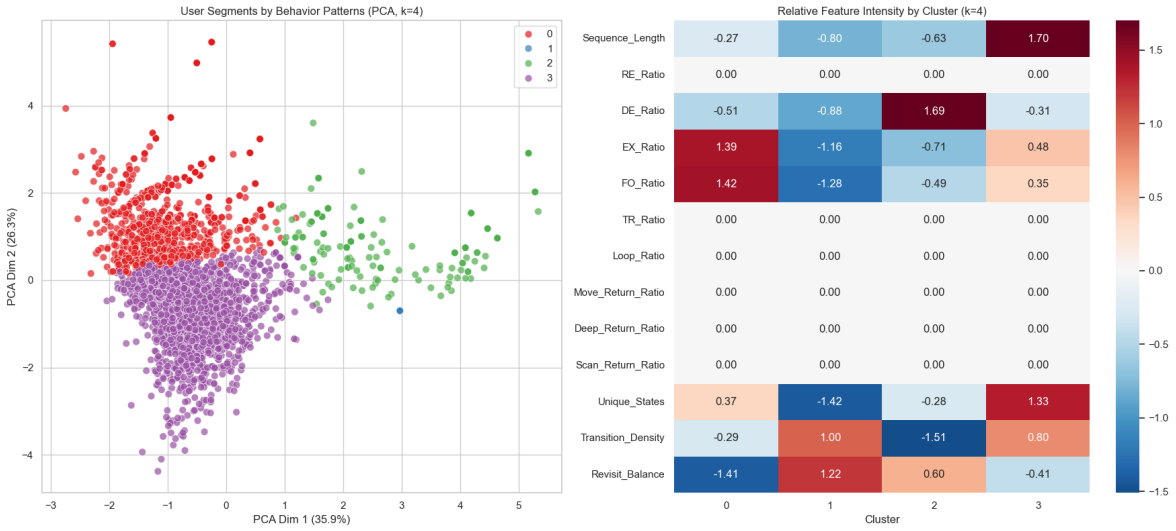


Figure 5 – Four-cluster solution: PCA projection (left) and standardised feature intensity by cluster (right)

Real stratum: demi-regularities and the Re-engagement Loop

Tuning granularity upward from k = 2 to k = 5 shows that two structures persist across all solutions: (a) a large, activity-rich exploratory-browsing cluster, and (b) a small, depth-oriented cluster. These are the demi-regularities that survive granularity tuning and that we therefore treat as candidate mechanisms. The exploratory-browsing mechanism is consistent with a spatial-affordance reading: zones form a circulation loop and the built environment rewards continuous forward movement with periodic returns to attractive anchors (the high-dwell zones of Figure 3). The Re-engagement Loop, operationalised as revisits separated by gaps beyond 180 seconds, is concentrated precisely in this cluster—the zones generating the heaviest-tailed dwell (9, 12, 6) are also the zones to which high-activity users most frequently return—suggesting that the Loop is less a demographic trait than an affordance-driven regularity of the service space.

Discussion

The empirical results give concrete content to the theoretical claim that ontological granularity can be tuned to surface demi-regularities. The exploratory-browsing routine is stable across granularities (k = 2–5) and is further supported by the zone-level evidence that a small set of anchor zones drive the heaviest dwell and the most re-engagement. This pattern is consistent with prior service-design arguments that experience environments are engineered to produce repetitive interactional sequences (Chase and Dasu, 2001; Zomerdijk and Voss, 2010). By contrast, the deep-engagement archetype is small and qualitatively different, suggesting that deep cognitive absorption is not merely an intensified version of browsing but a distinct mechanism.

For behavioural operations management, three implications follow. First, computationally intensive theorising benefits from an explicit granularity protocol: fixing granularity ex ante risks either averaging away the deep-engagement mechanism (coarse k) or fragmenting the dominant exploratory routine (fine k). Second, operationalising retroduction through CSA plus clustering provides a tractable pipeline for moving from digital traces to mechanism claims, extending Berente et al.'s (2019) call and sharpening Zhang et al.'s (2022) framework for service-oriented settings. Third, the affordance-driven stability of the exploratory-browsing routine suggests that operational interventions targeting layout and anchor-zone design will have more traction than demographic segmentation for shaping visitor engagement.

Managerially, the findings translate into concrete levers. Because roughly 54% of users follow the High-activity explorer routine, service designers should invest in the zones that function as anchors of this routine, widening entrances, easing sightlines and scheduling refresh events to convert mid-range browsers (27.4%) into higher-depth users. Because the Focused deep-engager routine is small but persistent, reserved spaces for sustained attention (seating, guided demonstrations, scheduled talks) are disproportionately valuable even though they serve a minority of traffic.

Conclusion, limitations and further work

This paper develops ontological granularity tuning as a critical realist method for computationally intensive theorising of service routines. Applied to a 2,871-visitor, 90,420-event exhibition dataset, it uncovers four routine archetypes, a dominant exploratory-browsing routine whose stability across granularities is interpreted as evidence of an underlying affordance-driven mechanism, and a small but distinctive focused deep-engagement routine. Three contributions follow: methodologically, we formalise granularity tuning and operationalise retroduction through CSA and clustering; empirically, we document demi-regularities in an experiential service; managerially, we translate the routine archetypes into layout-centred design levers.

Limitations include reliance on zone-level RFID traces (finer within-zone behaviours are not observed), a single-venue context, and the symmetric treatment of age bands in this paper (a planned extension will replicate the granularity-tuning protocol within age bands to test the affordance-over-demographics claim directly). Further work will expand the alphabet of behavioural blocks, test the Re-engagement Loop as an explanatory mechanism under simulated layout interventions, and compare experiential service contexts (e.g., museums, flagship retail, healthcare waiting environments) to establish the boundary conditions of the routine archetypes identified here.

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